

# TABLE WARS

Competition Manager

Comprehensive Site Description

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*The Ultimate Dinner Table Competition Platform*

Boarding House Entertainment Programme

5 Rounds | Dual-Screen Presentation | Real-Time Scoring

Version 2.0 | May 2026

## 1. Executive Overview

Table Wars Competition Manager is a purpose-built web application designed to run a high-energy, five-round dinner table competition for boarding house students. The competition determines which table earns the coveted first position in the food line for an entire week. Built as a single-page Next.js application with client-side state management, the platform delivers a professional, presentation-grade experience that rivals dedicated presentation software while remaining lightweight enough to run entirely in a browser without any server infrastructure beyond the initial page load.

The application is inspired by dual-screen presentation tools like Presenter by WorshipTools, featuring a dedicated operator control panel (Host Panel) on one screen and a separate audience-facing projection display (Scoreboard) on another. This architecture ensures that the game master retains full operational control over what the audience sees and when they see it, enabling dramatic reveals, timed sequences, and polished transitions that elevate the competition from a simple quiz game into a genuine event experience.

At its core, the system manages five distinct rounds of competition: Table Trivia (a buzzer-based quiz), Food Relay (a physical relay race), Table Building (a creative construction challenge), Blind Taste Test (a sensory identification challenge), and Grand Finale (a double-points showdown with a duel mechanic). Each round has unique scoring mechanics, interactive controls, and visual presentations. The application tracks scores in real time, animates rank changes, and delivers celebration effects when rounds conclude.

## 2. Architecture & Technical Stack

### 2.1 Core Technology

The application is built on Next.js 16 with React 19, leveraging the latest features of the React ecosystem for responsive, component-driven UI development. State management is handled by Zustand, a lightweight and performant state library that provides a single global store for all game state. This ensures that every view, whether the host panel, the scoreboard, or a team's mobile device, reads from the same source of truth and reacts to changes instantly. The use of Framer Motion for animations enables smooth, spring-based transitions, layout animations for rank changes, and celebration particle effects that give the application a polished, broadcast-quality feel.

The UI layer is built on shadcn/ui, a collection of beautifully designed, accessible components built on top of Radix UI primitives. This provides consistent design language across all views, from cards and buttons to dialogs and progress bars. Tailwind CSS 4 handles the styling layer, enabling rapid iteration on visual design while maintaining a

cohesive color system and spacing scale. The entire application is client-side rendered, meaning once the page loads, no further server communication is required to run the competition.

## **2.2 Sound System**

All sound effects are generated in real time using the Web Audio API, eliminating the need for audio file assets. The system produces six distinct sounds: a buzzer (ascending square-wave tones at 440, 554, and 659 Hz for urgency), a correct-answer chime (ascending sine-wave chord C5-E5-G5 for satisfaction), a wrong-answer buzz (descending sawtooth from 300 to 200 Hz for clarity), a timer alert ping (880 Hz sine wave for attention), a quiet tick sound (600 Hz sine, played during the last ten seconds of a countdown), and a celebration arpeggio (C5-E5-G5-C6 major chord for excitement). This approach ensures zero-latency audio response and eliminates the complexity of managing audio file loading and buffering.

## **2.3 Client-Side Architecture**

The application operates entirely on the client side after initial page load. Game state is stored in memory via Zustand and is not persisted to any backend. This design choice was intentional: it ensures simplicity, eliminates server dependency during the live event, and prevents any latency that could disrupt the real-time flow of the competition. The trade-off is that a page refresh during a live competition would reset all state; however, given that the event typically runs for 60-90 minutes with a dedicated laptop, this risk is minimal and manageable. A future enhancement could add localStorage or IndexedDB persistence for crash recovery.

# **3. Dual-Screen Presentation System**

One of the most distinctive features of the Table Wars Competition Manager is its dual-screen presentation architecture, directly inspired by professional presentation software like Presenter by WorshipTools. In a typical live-event setting, the game master operates the competition from a laptop (the operator station), while a projector or large display shows the audience-facing scoreboard (the audience display). The application provides separate, purpose-built views for each screen, with the operator retaining granular control over what appears on the audience display at all times.

## **3.1 Operator View (Host Panel)**

The Host Panel is the game master's command centre. It provides a comprehensive control interface that includes the current question or taste test item, buzz-in controls, scoring

buttons, timer controls, round management, and presentation mode switching. The left sidebar displays real-time team standings with animated rank changes, while the confidence monitor (a miniature preview of the audience display) shows exactly what is currently being projected. This confidence monitor is crucial for the game master to verify that the audience is seeing the intended content before making reveals or transitions.

The operator has access to a suite of keyboard shortcuts that enable rapid, hands-on control without reaching for the mouse. Pressing Space reveals the current question or taste test item; the right arrow key advances to the next question or item; T toggles the timer; B switches the projector to a black screen; L displays the animated logo screen; number keys 1 through 8 trigger buzz-ins for the corresponding team; S switches the current browser tab to the scoreboard view; H returns to the host panel; and Escape clears the current buzz. These shortcuts are designed so the game master can operate the entire competition from a keyboard with minimal visual distraction.

### 3.2 Audience Display (Scoreboard)

The Scoreboard is designed for projection on a large screen visible to the entire audience. It features a dark, cinematic colour scheme with animated leaderboards, live content displays for questions and taste test items, timer readouts, buzz-in indicators, and celebration particle effects. The display supports four distinct projector modes: Scoreboard (the default live competition view), Black (a pure black screen for hiding content between rounds or during transitions), Logo (an animated title screen with the TABLE WARS branding and pulsing effects), and Announcement (a custom text banner that scrolls across the top of the display). The game master can switch between these modes instantly using the Host Panel controls or keyboard shortcuts.

Round transitions are handled with dramatic splash overlays that announce the new round name with spring-based scale animations. These splashes auto-clear after three seconds, providing a moment of anticipation before the round begins. When a round concludes, celebration particles rain down across the display for three seconds, creating a visual reward for the teams and energizing the audience. The scoreboard also supports fullscreen mode via the Fullscreen API, allowing it to fill the entire projector display without browser chrome or operating system taskbars visible.

### 3.3 Confidence Monitor

Embedded within the Host Panel sidebar is a confidence monitor, a miniature real-time preview of what is currently displayed on the audience screen. This feature, directly borrowed from professional presentation tools, allows the game master to verify the current projector mode, confirm that questions are hidden before they are meant to be revealed, check timer status, and ensure announcements are displaying correctly. The confidence

monitor updates in real time alongside the actual scoreboard, so there is never a disconnect between what the operator expects and what the audience sees. This is particularly important during quiz rounds, where premature question reveals would compromise the competition, and during the Blind Taste Test, where revealing the identity of an item too early would spoil the challenge.

### **3.4 Black Screen and Logo Modes**

The Black Screen mode provides a clean, distraction-free visual state for the audience display. It is used between rounds to refocus attention on the game master's verbal instructions, during setup periods when teams are preparing for physical challenges, and during any pause in the competition where a visible scoreboard would be distracting. The Logo mode displays the TABLE WARS animated title with a pulsing scale effect on a dark purple gradient background, serving as a branded interstitial screen that maintains the event's visual identity during breaks and transitions. Both modes can be activated instantly via keyboard shortcuts (B for Black, L for Logo) and toggled back to the scoreboard with a single additional keypress.

### **3.5 Announcement System**

The Announcement mode enables the game master to display custom text banners overlaid on the scoreboard. This is useful for communicating housekeeping messages (such as break durations or schedule changes), encouraging audience participation, or adding dramatic flair (such as revealing the final standings heading). The announcement text appears as an animated banner at the top of the scoreboard with a gradient amber background, ensuring it is immediately visible without obscuring the leaderboard. The game master types the announcement in a text input on the Host Panel, and it appears on the scoreboard in real time as they type, with a subtle scaling animation that draws the eye.

## **4. Five-View System**

The application is organized into five distinct views, each serving a specific role in the competition lifecycle. Views are managed by the Zustand store's `currentView` state, and switching between them is instantaneous since all views share the same underlying state.

### **4.1 Setup View**

The Setup View is the landing page where the game master configures the competition before it begins. It features a dramatic hero section with the TABLE WARS title and tagline, followed by team configuration controls. The game master can set the number of teams (between 3 and 8), assign each team a name, choose a team colour from a palette of eight

vibrant colours, and set the member count. Team cards are arranged in a responsive grid, and each team's colour is visually indicated with a circle swatch and a tinted card border.

Below the team configuration, the Setup View provides a collapsible section for customizing the Blind Taste Test items (detailed in Section 6) and a collapsible rules summary that outlines all five rounds and their scoring mechanics. The start button is disabled until at least three teams are configured with non-empty names, preventing accidental starts with incomplete data. When the competition is started, the view transitions automatically to the Host Panel.

## 4.2 Host Panel

The Host Panel is the primary operational interface described in detail in Section 3.1. It adapts its main content area based on the current round, displaying quiz controls for Rounds 1 and 5, placement selectors for Rounds 2 and 3, taste test controls for Round 4, and duel mechanics for Round 5. The sidebar provides persistent access to standings, the confidence monitor, projector controls, and keyboard shortcut reference. Between rounds, the panel displays a transition screen with options to proceed to the next round or end the competition.

## 4.3 Live Scoreboard

The Live Scoreboard is the audience-facing display described in Section 3.2. It is designed to be opened in a separate browser tab or window and displayed on a projector. It receives all state updates from the shared Zustand store, ensuring perfect synchronization with the Host Panel. The scoreboard features animated leaderboards with score bars that fill proportionally, rank badges with team colours, round-by-round score breakdowns, and dramatic scale animations for the leading team.

## 4.4 Team View

The Team View is a mobile-optimized interface designed for team captains to use on their phones. Each team can access their own dashboard, which displays their current score, rank, and round-by-round breakdown. Most importantly, the Team View includes a large buzz-in button that teams can press during quiz rounds. The button uses touch-optimized sizing and provides immediate visual and haptic feedback when pressed. Once a team has buzzed in, the button disables to prevent double-presses, and the team's name appears on the scoreboard as the buzzed-in team. The Team View is accessed by navigating to the application URL on a mobile device and selecting the appropriate team.

## 4.5 Results View

The Results View is the grand finale of the application, displaying the final standings with a progressive reveal animation. Teams are shown from last place to first, with each reveal building anticipation. The winning team receives a special crown animation and confetti celebration. Below the standings, the view provides a sportsmanship voting section where each team can vote for another team that demonstrated the best sportsmanship throughout the competition. This voting mechanism promotes positive behaviour and ensures that the competition culture remains supportive and fun. An awards section displays the champion, runner-up, and sportsmanship award recipient.

## 5. Five Rounds of Competition

### 5.1 Round 1: Table Trivia

Table Trivia is a buzzer-based quiz round featuring 20 multiple-choice questions spanning categories including Science, Geography, Art, Math, History, Food, Literature, Sports, Nature, Technology, and Pop Culture. The game master controls the pace by revealing questions one at a time. When a question is revealed, teams race to buzz in first. The first team to buzz in gets the opportunity to answer; if they answer correctly, they earn 5 points. If they answer incorrectly, the remaining teams have an opportunity to steal the question for 3 points. This steal mechanic ensures that every question remains engaging even after an incorrect answer, as teams stay alert for the chance to capitalize on their opponents' mistakes.

The Host Panel provides a question card that displays the current question text, answer options, and correct answer (visible only to the operator). Below the card, a grid of team buzz-in buttons allows the operator to register which team buzzed in, either by clicking the team's button or pressing the corresponding number key. Once a team is registered as buzzed in, two action buttons appear: Correct (+5) and Wrong. If the answer is wrong, steal buttons appear for all other teams. A quick-jump grid of numbered buttons allows the operator to skip to any question out of order if needed, and the question card includes a progress indicator showing the current question number out of the total.

### 5.2 Round 2: Food Relay

The Food Relay is a physical relay race round where teams compete head-to-head. Rather than a buzzer or quiz mechanic, this round uses a placement-based scoring system. The game master observes the race and assigns each team a finishing placement (1st through 8th). Placement points are awarded on a descending scale: 1st place receives 25 points, 2nd receives 20, 3rd receives 15, 4th receives 10, 5th receives 5, 6th receives 3, 7th receives 2, and 8th receives 1 point. Additionally, a penalty of minus 3 points is applied for each dropped item during the relay, encouraging teams to balance speed with care.

The Host Panel for Round 2 provides a dropdown selector for each team's placement and increment/decrement buttons for penalties. The operator assigns placements as the race unfolds and records any penalties. Once all teams have been assigned a placement, the Lock Round 2 Scores button becomes active. Locking the round finalizes all scores, calculates the net points (placement points minus penalties), and writes them to the team records. After locking, the placement selectors become disabled to prevent accidental changes.

### **5.3 Round 3: Table Building**

The Table Building round challenges teams to construct the tallest possible structure using provided materials. Like the Food Relay, it uses a placement-based scoring system with the same point scale (1st = 25 points down to 8th = 1 point). Additionally, a creativity bonus of 5 points can be awarded to any team at the game master's discretion, adding a subjective element that rewards innovation and design thinking alongside raw height. The Host Panel provides placement selectors and a creativity bonus toggle for each team. When the creativity toggle is activated, a visual indicator highlights the bonus on the team's row, and the scoreboard reflects the additional points in real time.

### **5.4 Round 4: Blind Taste Test**

The Blind Taste Test is a sensory identification challenge where teams attempt to identify mystery food items while blindfolded. The game master reveals one item at a time, and each team earns points based on their accuracy: 4 points for a fully correct identification, 2 points for a partial identification (correct category but wrong specific item), and 0 points for an incorrect answer. The Host Panel provides a taste test card showing the current item's hint and category, along with a grid of scoring buttons (Correct, Partial, Wrong) for each team. A bug-fix mechanism tracks per-team per-item scores to prevent double-counting when the operator changes a team's score for the same item.

This round is where the application's presentation-inspired architecture truly shines. The game master controls the reveal of each taste test item using the Space key (matching the quiz round's reveal mechanic), and the audience sees only the item's hint and category on the scoreboard, never the actual item name. This creates a shared mystery experience where the entire audience is guessing along with the teams. The operator can advance through items sequentially or jump to specific items, and the confidence monitor ensures that the operator always knows exactly what the audience is currently seeing.

### **5.5 Round 5: Grand Finale**

The Grand Finale is the culminating round where all points are doubled, making it the most consequential segment of the competition. It features 15 higher-difficulty questions (compared to Round 1's 20 general-knowledge questions), with correct answers earning 10



points instead of the usual 5. The steal mechanic remains, but even steals are worth more in this round. The increased point values mean that the leaderboard can shift dramatically, ensuring that no team's position is secure until the very last question.

In addition to the quiz format, Round 5 introduces a Duel mechanic. The game master can activate duel mode, which pits two selected teams against each other in a head-to-head showdown. The duel challenger and challenged teams are selected via the Host Panel, and the winner receives an additional 10-point bonus. This mechanic adds strategic depth, as teams must decide when to deploy their duel opportunity and against which opponent. The duel is visually represented on the scoreboard with crossed-swords indicators for the duelling teams.

## 6. Customizable Content System

The Table Wars Competition Manager provides a comprehensive content customization system that allows the game master to tailor every aspect of the competition's content to their specific audience, venue, and available materials. This system covers both the Blind Taste Test items and the quiz question banks, ensuring that no competition is ever identical and that the game master has full creative control.

### 6.1 Customizable Taste Test Items

The Blind Taste Test round features a fully customizable item system. The application ships with 10 default taste test items (Soy Sauce, Honey, Lemon Juice, Olive Oil, Peanut Butter, Coconut Milk, Maple Syrup, Balsamic Vinegar, Marmite, and Cream Cheese), each with a descriptive hint and a category label. However, the game master can modify every aspect of these items before or during the competition.

The customization interface is available in two locations: the Setup View (pre-competition) and the Host Panel (mid-competition, via an Edit Items dialog). In both locations, the game master can:

- **Add new items** by providing an item name, a descriptive hint, and a category. New items are assigned unique IDs and appended to the end of the item list.
- **Edit existing items** by clicking the pencil icon next to any item. The item name, hint, and category become editable inline, and changes are saved with a single click.
- **Remove items** by clicking the trash icon. Removed items are immediately excluded from the rotation, and the item count updates accordingly.
- **Reorder items** using up/down arrow buttons. The order determines the sequence in which items are presented during the round, allowing the game master to build dramatic pacing.

- **Reset to defaults** with a single button that restores the original 10-item set, useful if the operator has made unwanted changes.

A badge indicator on the Setup View marks the taste test section as Customized when the game master has made any changes, providing a visual cue that the default set has been modified. During the competition, the Host Panel's Edit Items dialog provides the same full editing capabilities, enabling mid-round adjustments if, for example, a particular food item is unavailable on the day.

## 6.2 Customizable Quiz Questions and Question Banks

The quiz rounds (Round 1: Table Trivia and Round 5: Grand Finale) feature a flexible question management system that allows the game master to either use the built-in question banks or supply their own questions. The application ships with 35 pre-loaded questions: 20 general-knowledge questions for Round 1 and 15 higher-difficulty questions for Round 5. These questions span diverse categories including Science, Geography, Art, Math, History, Food, Literature, Sports, Nature, Technology, Pop Culture, and Music.

The question customization system is designed around two complementary approaches:

### 6.2.1 Built-In Question Banks

The default question banks provide a ready-to-play experience that covers a broad range of topics suitable for a boarding house audience of mixed ages and interests. Each question includes a unique ID, the question text, a correct answer, a category label, and four multiple-choice options. The questions are ordered by design, with the Round 1 set progressing through categories to maintain variety and the Round 5 set featuring intentionally harder material (such as identifying Olympus Mons as the tallest mountain in the solar system or recognizing that the Antarctic Desert is the world's largest desert). These banks can be used as-is for a complete, balanced competition experience.

### 6.2.2 Custom Question Editor

For game masters who want to tailor the quiz content to their specific audience, the application provides a question editor accessible from both the Setup View and the Host Panel. The editor allows the game master to:

- **Add new questions** by entering the question text, correct answer, category, and four multiple-choice options. New questions are validated to ensure all fields are populated before being added to the bank.
- **Edit existing questions** by selecting any question from the bank and modifying its text, answer, category, or options. Changes are applied immediately and reflected in the next play-through of that round.

- **Remove questions** that are not appropriate for the audience or are redundant. The minimum question count is enforced to ensure there are enough questions to complete the round.
- **Import question sets** from a structured format (such as JSON or CSV), enabling game masters to prepare questions in advance on a spreadsheet and bulk-import them. This is particularly useful for themed competitions (e.g., a holiday-themed quiz or a subject-specific academic challenge).
- **Reorder questions** to control the pacing and difficulty curve within each round. The game master can front-load easier questions and build toward more challenging ones, or interleave categories for variety.
- **Reset to default bank** to restore the original question set if customizations need to be discarded.

The question editor interface mirrors the taste test editor's design, with inline editing, drag-to-reorder functionality, and a Reset to Defaults button. Questions are displayed in a scrollable list with category badges and difficulty indicators, allowing the game master to quickly scan and adjust the question set. During the competition, the editor can be accessed via a dialog from the Host Panel, enabling real-time adjustments such as removing a question that was accidentally revealed or substituting a question that is not working well with the audience.

### 6.2.3 Question Bank Architecture

The question management system is built on a flexible architecture that separates the data layer from the presentation layer. Questions are stored as an array of QuizQuestion objects in the Zustand store, with each object containing an ID, question text, answer, category, and optional array of multiple-choice options. When a round begins, the relevant question bank (Round 1 or Round 5) is loaded into the store's questions array. The game master can modify this array at any time using the store's question management actions, which include adding, editing, removing, and reordering questions, as well as bulk-replacing the entire question set.

This architecture supports future extensibility, such as:

- Multiple themed question packs (e.g., Sports Pack, Science Pack, Pop Culture Pack) that can be selected or combined before the competition.
- Difficulty tagging (Easy, Medium, Hard) that enables the game master to filter questions by difficulty or configure automatic difficulty progression.
- Audience-submitted questions, where teams can submit questions in advance and the game master can curate a selection from those submissions.

- Random question selection from a larger pool, where the game master specifies the number of questions to play and the system randomly selects from the bank, ensuring a different experience each time.

## 7. Scoring & Rankings

### 7.1 Scoring Mechanics

Each round employs a distinct scoring mechanism tailored to its format. The following table summarizes the scoring rules for each round:

| Round | Name             | Scoring Rule                                     | Max Points (per team) |
|-------|------------------|--------------------------------------------------|-----------------------|
| 1     | Table Trivia     | Correct: +5, Steal: +3                           | 100 (20 x 5)          |
| 2     | Food Relay       | Placement: 25/20/15/10/5/3/2/1, Penalty: -3/drop | 25 + bonuses          |
| 3     | Table Building   | Placement: 25/20/15/10/5/3/2/1, Creativity: +5   | 30 (25 + 5)           |
| 4     | Blind Taste Test | Correct: +4, Partial: +2                         | 40 (10 x 4)           |
| 5     | Grand Finale     | Correct: +10, Steal: +3, Duel Win: +10           | 150+                  |

### 7.2 Real-Time Leaderboard

The leaderboard is the centrepiece of the Scoreboard display. It updates in real time as scores are added, with animated transitions that visually communicate rank changes. Teams are sorted by total score in descending order, and each team's entry displays their rank badge (coloured circle with the team's colour), team name, round-by-round score breakdown (small badges showing R1, R2, R3, R4, R5 scores), and total score in large, bold type. The first-place team receives a special visual treatment: a crown emoji instead of a numeric rank, a glowing box-shadow in the team's colour, and a subtle pulsing scale animation that draws attention.

Score changes are animated with spring-based transitions: when a team's score increases, the total score number briefly scales up and flashes amber before settling back to its normal size. Rank changes are animated with layout animations, so teams physically slide up or down the leaderboard as their positions shift. This creates a dynamic, engaging visual experience that keeps the audience invested in the standings throughout the competition.

### 7.3 Score Correction and Double-Counting Prevention

The scoring system includes safeguards against common operator errors. For the Blind Taste Test round, a tracking mechanism (`tasteItemScores`) records the specific score awarded to each team for each item. If the operator changes a team's score for the same item (for example, upgrading from Partial to Correct), the system automatically subtracts the previous score before adding the new one, preventing double-counting. For placement-based rounds, the Lock mechanism finalizes scores and disables the input controls, preventing accidental post-lock changes. For quiz rounds, each correct or steal answer is recorded as a separate score entry with a descriptive detail string, providing an audit trail that the operator can review if questions arise about a team's score.

## 8. Timer & Buzzer Systems

### 8.1 Countdown Timer

The integrated countdown timer supports preset durations of 30, 60, 90, and 120 seconds, with the default set to 60 seconds. The timer is controlled via the Host Panel with start, stop, and reset buttons, as well as the T keyboard shortcut for quick toggling. When the timer is running, it displays on both the Host Panel and the Scoreboard, with colour-coded urgency: the display turns amber when 10 seconds remain and red when 5 seconds remain, creating a visual countdown that builds tension. During the final 10 seconds, a quiet tick sound is played each second via the Web Audio API, and a distinct alert ping sounds when the timer reaches zero.

### 8.2 Buzz-In System

The buzz-in system is the primary interaction mechanism for quiz rounds (Rounds 1 and 5). Teams can buzz in via two channels: the Host Panel operator can press number keys 1-8 or click the corresponding team button, and team captains can press the buzz button on their mobile Team View. When a team buzzes in, the buzzer sound plays, the team's button visually locks with a highlighted state, and all other teams' buttons become disabled. The scoreboard displays a prominent buzz-in indicator showing the team's name. The operator then adjudicates the answer using the Correct or Wrong buttons, or offers steal opportunities to the remaining teams if the answer was incorrect. The buzz can be cleared at any time using the Escape key or the Clear Buzz button.

## 9. Presentation Features (WorshipTools-Inspired)

The Table Wars Competition Manager incorporates several design principles and features inspired by professional presentation software, particularly Presenter by WorshipTools. These features elevate the competition from a simple scoring application into

a full production tool that gives the game master broadcast-level control over the audience experience.

## **9.1 Dual-Screen Operator/Audience Architecture**

Like Presenter by WorshipTools, the application is built around a dual-screen paradigm where the operator (game master) and audience see different views on different displays. The Host Panel serves as the operator console, providing full editorial control over content, timing, and transitions, while the Scoreboard serves as the audience projection, displaying only the content that the operator has chosen to reveal. This separation ensures that the operator can prepare content, review answers, and set up transitions without the audience seeing behind-the-scenes activity. The confidence monitor within the Host Panel provides a real-time miniature preview of the audience display, mirroring the stage display feature in WorshipTools.

## **9.2 Slide-Based Content Progression**

Questions and taste test items function like slides in a presentation deck. Each item is a discrete content unit that can be hidden or revealed independently, and the operator advances through them sequentially (or jumps to specific items using the quick-jump grid). The reveal mechanic (Space key) mirrors the action of advancing a slide in presentation software, showing the next piece of content to the audience while keeping the answer or item name hidden on the operator's screen. This slide-based approach gives the operator precise control over pacing and dramatic timing, enabling them to build suspense before a reveal or pause for emphasis after a particularly challenging question.

## **9.3 Media Display and Reveal Controls**

The reveal system acts as a media display controller, managing what content is visible on the audience screen at any given time. Questions are initially hidden (showing a locked indicator on the scoreboard), and the operator controls when they become visible. For the taste test, only the hint and category are shown to the audience; the actual item name is visible only on the Host Panel, preserving the mystery. This controlled reveal mechanism is analogous to the media cue system in presentation software, where operators queue up content and trigger it at the optimal moment.

## **9.4 Transitions and Visual Effects**

Content transitions between questions and taste test items use Framer Motion spring-based animations with horizontal slide effects, creating a smooth, cinematic flow between content units. Round transitions feature dramatic splash overlays with scale animations that announce the new round name. Celebration effects (colourful falling particles) are triggered

when rounds conclude, providing visual punctuation marks that signal transitions in the competition narrative. These transitions are designed to be dramatic enough to create a sense of occasion while being fast enough not to interrupt the competition's momentum.

## **9.5 Blank/Black Screen Control**

The Black Screen mode (activated by pressing B) is directly equivalent to the blank screen feature in presentation software, which operators use to temporarily hide projected content and refocus the audience's attention on the speaker. In the Table Wars context, the game master uses Black Screen between rounds, during setup periods for physical challenges, and whenever the scoreboard would be distracting or premature. The transition to and from Black Screen is instantaneous, ensuring the operator can react quickly to unexpected situations or timing changes.

## **9.6 Stage Display (Confidence Monitor)**

The confidence monitor is the Table Wars equivalent of WorshipTools' stage display feature, which shows performers a preview of what the audience is seeing along with additional operator-only information. In the Table Wars application, the confidence monitor displays the current projector mode, the current question or taste test item (with hidden answers shown as a locked indicator), the timer status, and the round number. This allows the game master to verify the audience display state at a glance without needing to look at the projector screen, which may be positioned behind them or in a different part of the room.

## **9.7 Announcement Overlay**

The announcement overlay is inspired by the lower-third and banner features in presentation software, which display text messages over the main content without interrupting it. The game master can type a message in the Host Panel and switch to Announcement mode, causing the message to appear as an animated amber banner at the top of the scoreboard. This is useful for housekeeping announcements, audience instructions, or dramatic reveals such as announcing the prize or final scores. The announcement coexists with the leaderboard, so the audience can continue to see the standings while reading the message.

# **10. Device Strategy**

The application is designed for a multi-device deployment during a live event. The recommended setup involves three categories of devices, each serving a specific purpose:

## **10.1 Operator Laptop (Host Panel)**

The game master uses a laptop as the primary control station, running the Host Panel view. This device requires a keyboard for shortcut-based operation and a screen large enough to display the sidebar standings, confidence monitor, and main content area simultaneously. The laptop should be positioned where the game master can see the projector screen (or use the confidence monitor as a proxy) while maintaining eye contact with the audience and teams.

## 10.2 Projector or Large Display (Scoreboard)

A projector or large-format display shows the Scoreboard view to the entire audience. This display should be positioned centrally in the dining hall, visible from all tables. The scoreboard is opened in a separate browser tab and switched to fullscreen mode using the Fullscreen API. The game master navigates to the scoreboard view from the Host Panel (or by pressing S), and then moves that tab to the projector display. The two tabs share the same Zustand store, ensuring perfect synchronization.

## 10.3 Team Captain Mobile Devices (Team View)

Each team captain uses a mobile phone to access the Team View, which provides their personal dashboard with score information and a buzz-in button. The phones connect to the same application URL in their mobile browser, and each team selects their identity to access their specific dashboard. The buzz-in button is touch-optimized with large hit targets and immediate visual feedback, making it easy to press quickly during quiz rounds. The responsive design ensures the interface works well on screens from 320px to 428px wide, covering the full range of modern smartphone sizes.

# 11. Keyboard Shortcuts Reference

The following table summarizes all keyboard shortcuts available to the game master during the competition:

| Key         | Action                            | Context        |
|-------------|-----------------------------------|----------------|
| Space       | Reveal question / taste test item | Rounds 1, 4, 5 |
| Right Arrow | Next question / taste test item   | Rounds 1, 4, 5 |
| T           | Toggle timer start/stop           | All rounds     |
| B           | Toggle black screen on projector  | Any time       |
| L           | Display logo screen on projector  | Any time       |
| 1-8         | Buzz in team 1-8                  | Rounds 1, 5    |
| S           | Switch current tab to scoreboard  | Any time       |



| Key    | Action                           | Context     |
|--------|----------------------------------|-------------|
| H      | Switch current tab to host panel | Any time    |
| Escape | Clear current buzz               | Rounds 1, 5 |

## 12. Default Data Summary

### 12.1 Default Taste Test Items

| Item             | Hint                                               | Category          |
|------------------|----------------------------------------------------|-------------------|
| Soy Sauce        | A dark, salty liquid used in Asian cuisine         | Condiment         |
| Honey            | A golden sweet substance made by bees              | Sweetener         |
| Lemon Juice      | A tangy citrus liquid                              | Citrus            |
| Olive Oil        | A smooth, rich liquid from pressed olives          | Cooking Oil       |
| Peanut Butter    | A thick, nutty spread                              | Spread            |
| Coconut Milk     | A creamy white liquid from a tropical fruit        | Dairy Alternative |
| Maple Syrup      | A sweet amber liquid from tree sap                 | Sweetener         |
| Balsamic Vinegar | A dark, sweet-and-sour Italian condiment           | Condiment         |
| Marmite          | A sticky, dark spread with a strong savory flavour | Spread            |
| Cream Cheese     | A soft, mild, white spreadable cheese              | Dairy             |

### 12.2 Question Bank Summary

| Round            | Questions | Categories Covered                                                                                |
|------------------|-----------|---------------------------------------------------------------------------------------------------|
| 1 (Table Trivia) | 20        | Science, Geography, Art, Math, History, Food, Literature, Sports, Nature, Technology, Pop Culture |
| 5 (Grand Finale) | 15        | Nature, Technology, Science, Music, Geography, Literature, Art, History                           |

All questions are multiple-choice with four options and a single correct answer. The Round 1 questions are designed for general accessibility (e.g., identifying gold's chemical symbol as Au, naming the Red Planet as Mars), while Round 5 questions require deeper knowledge (e.g., identifying Olympus Mons as the tallest mountain in the solar system, knowing that Marie Curie was the first woman to win a Nobel Prize).

## 13. Future Enhancements

While the current application provides a complete, production-ready competition management experience, several enhancements are planned for future versions that would further elevate the platform's capabilities:

- **WebSocket-based multi-device synchronization:** Currently, all views share state through the same browser's Zustand store, requiring the scoreboard to be in a separate tab on the same device. WebSocket support would enable true multi-device operation, where the scoreboard runs on a different computer connected to the projector, and team captain phones connect over WiFi.
- **Persistent state via localStorage:** Adding automatic state serialization to localStorage would protect against accidental page refreshes during a live competition, enabling crash recovery and the ability to resume an interrupted event.
- **Image and video media support:** Extending the slide-based content system to support image and video reveals would add a multimedia dimension to the quiz rounds, enabling picture-based questions and video clues for the taste test round.
- **Themed question packs:** A marketplace or import system for themed question packs (Holiday, Sports, Academic, Pop Culture) would allow game masters to quickly customize the quiz content for different occasions and audiences without manually editing individual questions.
- **Spectator mode:** A read-only spectator view optimized for large screens at home or in common areas, displaying the scoreboard with additional context such as question text, team statistics, and a live event timeline.
- **Advanced presentation transitions:** Adding configurable transition effects (fade, dissolve, zoom, slide) between content units, similar to the transition options available in professional presentation software, would give game masters more creative control over the visual narrative of the competition.